Mathematics

Teaching and Learning Syllabus

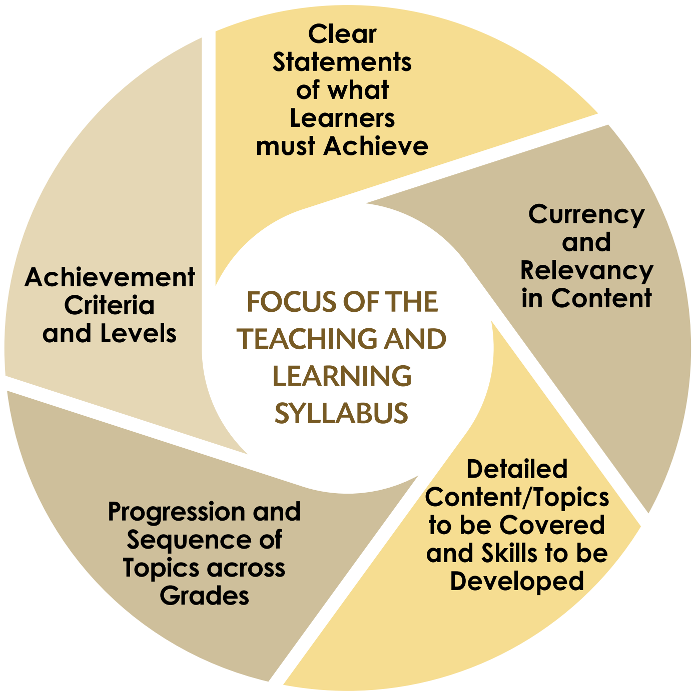
Grades 1 to 6

Mauritius Institute of Education
under the aegis of
Ministry of Education and Human Resources, Tertiary Education and Scientific Research

Teaching and Learning Syllabus
Mathematics

The Teaching and Learning Syllabus refers to the content of what is to be taught, and to the knowledge, skills, competencies, attitudes and values that are to be developed. It articulates concisely the specific learning outcomes for each grade as per the NCF 2015 and features achievable learning outcomes that represent a Teaching and Learning road map.

- **The focus of the TLS**



- **Clear statements of what learners must achieve**

The TLS spells out the specific learning outcomes that learners are expected to achieve progressively in terms of knowledge, understanding, skills, 21st Century Competencies, attitudes and values during each year of schooling, from Grades 1 to 6.

- **Currency and relevancy in content**

The TLS ensures currency and relevance of content and acquisition of developmentally-appropriate concepts and skills.

- **Detailed content/topics to be covered and skills to be developed**

The TLS details the content and topics to be covered as well as the skills to be developed for each grade while addressing the problem of curriculum overload and stress.

• Progression and sequence of topics across grades

The TLS outlines the scope and sequence of the specific learning outcomes for each grade. It indicates the grade in which knowledge, concepts, skills and processes related to the learning outcomes are introduced, based on a spiral curriculum. The learning outcomes are progressively sequenced across grades.

• Achievement criteria and levels

The TLS defines assessment criteria that are scaffolded and represent clear measurable outcomes. The levels reflect the competencies in terms of knowledge, skills, values and attitudes expected to have been acquired by learners. The development of achievement levels are meant to assist in the differentiation of teaching, learning and assessment so that the curriculum is accessible to all learners.

Contents

RATIONALE

Recent reforms in Mathematics Education and recommendations from the 'Deloading Document' have called upon a review in the teaching and learning of Mathematics. However, primary pupils need mathematical knowledge, skills and abilities in computation, logical thinking and problem solving to construct knowledge and understanding of the world around them. The pedagogical approaches for the teaching and learning of key mathematical concepts for Grades 1 to 6 have to be in line with current practices. Subsequently, an up-to-date Teaching and Learning Syllabus is warranted. The learning outcomes in this document describe the specific learning goals that are expected through Grades 1 to 6. It gives both broad and deep coverage of the content at the primary level. The Mathematics Teaching and Learning Syllabus also guides educators and examiners to prepare for teaching and learning as well as for formative and summative assessment for Grades 1 to 6.

EXPECTED LEARNING OUTCOMES IN MATHEMATICS

By the end of Standard 6, learners will be able to:

- Identify and manipulate numbers and perform number operations
- Use numbers in real life situations
- Use geometry and develop spatial abilities
- Use tools and appropriate units for measurement
- Organise, represent and analyse data
- Communicate effectively using mathematical vocabulary
- Demonstrate logical thinking
- Perform mental arithmetic/calculations
- Use patterns and make deductions
- Relate mathematics to other disciplines
- Use mathematical concepts learnt to formulate and solve mathematical problems
- Use ICT to support their learning and
- Appreciate the use of mathematics in daily life.

THE DEVELOPMENT OF 21ST CENTURY COMPETENCIES AND MATHEMATICS

An adequate knowledge of Mathematics is crucial to acquire the 21st Century Competencies as outlined in the National Curriculum Framework. The latest trends in mathematics education and the innovations in mathematics teaching and learning have shaped the development of the mathematics curriculum, where the 21st Century Competencies are embedded. The current emphasis of the mathematics curriculum for primary education is on developing mathematical proficiency for ALL learners. It advocates the use of the concrete-pictorial-abstract strategy, (for the development of concepts) which incorporates activity-based teaching strategies that integrates ICT. Components of mathematical proficiency include logical thinking, modelling, procedural fluency, conceptual understanding, communication, representation, efficient problem solving and strategic thinking. These eight components of the mathematical proficiency are at the core of the development of thinking, learning, communicative, civic as well as personal and social competencies for the 21st century learners. Thus, a sound knowledge of Mathematics enables a learner to become an informed citizen of society as well as escalate his/her chance to be successful in facing the challenges of the fast-changing world.

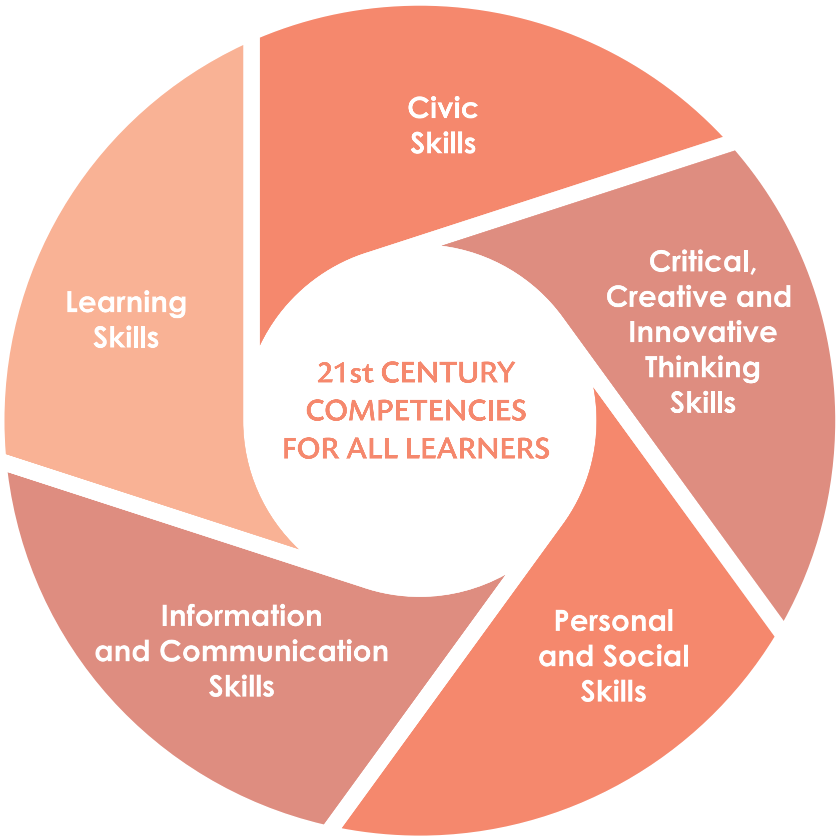


Figure 1: 21st Century Competencies for All Learners

SCOPE AND SEQUENCE OF CONTENT AREAS FOR MATHEMATICS

Content Areas	Grades					
	1	2	3	4	5	6
Numbers						
Whole Numbers	•	•	•	•	•	•
Numerals and Number Names	•	•	•	•	•	•
Place value		•	•	•	•	•
Ordinal Numbers	•	•	•			
Addition						
· Without Carrying	•	•	•	•	•	•
· With carrying			•	•	•	•
Subtraction						
· Without borrowing		•	•	•	•	•
· With borrowing			•	•	•	•
Multiplication		•	•	•	•	•
Division		•	•	•	•	•
Types of Numbers				•	•	•
Number Patterns		•	•	•	•	•
Fractions			•	•	•	•
Decimals					•	•
Factors						•
Multiples						•
HCF						•
LCM						•
Average					•	•
Ratio					•	•

Proportion					•	•
Powers					•	•
Percentage						•
Geometry						
2-D Shapes	•	•	•	•	•	•
3-D shapes				•	•	•
Lines				•	•	•
Angles				•	•	•
Symmetry					•	•
Measure						
Length	•	•	•	•	•	•
Mass	•	•	•	•	•	•
Time	•	•	•	•	•	•
Capacity		•	•	•	•	•
Money	•	•	•	•	•	•
Perimeter					•	•
Area					•	•
Volume					•	•
Speed						•
Graphs						
Pictograms			•	•	•	•
Bar charts				•	•	•
Pie charts						•
Line graphs						•
Coordinates						•

CONTENT AREAS AND SPECIFIC LEARNING OUTCOMES OF MATHEMATICS FOR GRADE 1

Content Areas	Topics/Sub-topics	By the end of Grade 1, learners should be able to:
Numbers	**Whole numbers** ***Numerals and Number Names (0-10)*** ***Ordinal Numbers (up to fifth)*** ***Addition (sum not exceeding 10)*** ***Number Patterns***	• Recognise, read and write numbers 0 to 10 in numerals and in words • Count and write numerals in a group (up to 10) • Match numerals to a number of items in a group and vice versa • Recognise and use ordinal numbers, and name the position of a member in a queue up to 5th • Perform addition of two numbers (sum not exceeding 10) • Recognise and complete number patterns • Perform and state orally simple mental calculations involving addition • Solve simple mathematical task involving addition of two numbers in context (sum not exceeding 10) • Write numbers from 0 to 10 and perform addition (sum not exceeding 10) using ICT
Shapes	**2-D Shapes** ***(triangle, rectangle, square and circle)*** **Sorting** ***(one attribute: colour, shape and size)*** ***Patterns with 2-D shapes***	• Recognise and name colours and shapes • Draw and name 2-D shapes (e.g., Triangle, Rectangle, Square, Circle) • Identify and compare shapes in the environment • Sort out objects according to one attribute and explain the sorting rule (colour, size and shape) • Identify and complete patterns using 2-D shapes • Draw and manipulate 2-D shapes using ICT
Measure	**Length** ***(long and short)*** **Mass** ***(heavy and light)*** **Money** ***(coins up to Rs 10)*** **Time** ***(morning/afternoon/day/night)***	• Sort and compare objects according to length • Sort and compare objects according to mass • Recognise and name coins up to Rs 10 • Decompose coins up to Rs 10 • Distinguish between time concepts (e.g., morning/ afternoon; day/night)

CONTENT AREAS AND SPECIFIC LEARNING OUTCOMES OF MATHEMATICS FOR GRADE 2

Content Areas	Topics/Sub-topics	By the end of Grade 2, learners should be able to:
Numbers	**Whole Numbers**	• Recognise, read and write numbers 0 to 100 in numerals and in words
	Numerals and Number Names (0-100)	• Count, compare and order numbers up to 100
	Place value (units, tens and hundred)	• Illustrate the meaning of place value for numerals up to 100
	Ordinal numbers (up to twelfth)	• Recognise and use ordinal numbers and name the position of a member in a queue up to 12th
	Addition (sum not exceeding 100; no carrying)	• Perform addition of two numbers (not involving carrying and sum not exceeding 100)
	Subtraction (no borrowing)	• Perform subtraction of two numbers (not involving borrowing)
	Multiplication (0 to 10 by 2)	• Multiply a number (from 0 to 10) by 2 (perform multiplication as a repeated addition, with and without manipulatives)
	Division (numbers up to 20 by 2, no remainder)	• Divide numbers up to 20 by 2
	Number patterns	• Recognise and complete patterns in number sequences
		• Perform and explain simple mental calculations involving addition and subtraction
		• Solve one-step word problems (addition, subtraction, multiplication and division)
		• Write numbers from 0 to 100 and perform the 4 arithmetic operations using ICT
Geometry	**2-D Shapes - Sorting** ***(two attributes: colour, shape and size)***	• Sort out objects according to two attributes and explain the sorting rule
	Patterns with 2-D Shapes	• Identify and complete patterns using 2-D shapes
		• Draw and manipulate basic 2-D shapes using ICT
Measure	**Length** ***(Arbitrary units)***	• Compare, estimate and measure length of objects using arbitrary units
	Mass ***(Arbitrary units)***	• Compare, estimate and measure mass of objects using arbitrary units
	Capacity ***(Arbitrary units)***	• Compare, estimate and measure capacity of objects using arbitrary units
	Money ***(Notes and coins up to Rs 100)***	• Recognise and name coins and notes up to Rs 100
		• Decompose coins and notes up to Rs 100
		• Use coins and notes up to Rs 100 in shopping situations
	Time ***(To the hour; Days, months)***	• Tell time to the hour from a clock face
		• Name days of the week and months of the year

CONTENT AREAS AND SPECIFIC LEARNING OUTCOMES OF MATHEMATICS FOR GRADE 3

Content Areas	Topics/Sub-topics	By the end of Grade 3, learners should be able to:
Number	**Whole numbers** ***Numerals and number names (0-1000)*** **Place value** ***(Units, tens, hundreds, thousands)*** **Addition** ***(up to 3 digits with and without carrying)*** **Subtraction** ***(up to 3-digits with and without borrowing)*** **Multiplication** ***(from 0-10 by 2, 3, 4 and 5)*** **Division** ***(multiples of 2, 3, 4 and 5 by 2, 3, 4 and 5)*** ***Number patterns*** **Ordinal Numbers** ***(Up to 20th)*** **Roman Numerals** ***(Up to 20)*** **Fractions** ***(Half, third, quarter)***	• Recognise, read and write the numbers up to 1000 in numerals and in words • Count, compare and order numbers up to 1000 • Illustrate the meaning of place value up to 1000 • Add 2 numbers (up to 3 digits) with and without carrying • Subtract 2 numbers (up to 3 digits) with and without borrowing • Multiply a number (from 0 to 10) by 2, 3, 4 and 5 • Divide a number (multiple of 2, 3, 4 and 5) by 2, 3, 4 and 5 respectively • Solve one-step word problems involving the 4 operations • Use numbers in real-life situations and solve both routine and non-routine word problems in real-life situations • Perform and explain mental calculations involving the 4 arithmetic operations • Recognise patterns in number sequences up to 1000 • Recognise and use ordinal numbers (1st to 20th) and name the position of a member in a queue up to 20th • Recognise, read and write roman numerals up to 20 • Use unit fractions as part of a whole (e.g. '1 out of 4' to describe a fraction) • Read, write and represent simple fractions (e.g. ½, 1/3, ¼) • Use fractions in real-life situations • Write numbers from 1 to 1000 and perform the 4 arithmetic operations using ICT
Shapes	**2-D Shapes** ***(Rectangle, Square,Circle, Triangle)***	• Manipulate, sort, represent, describe and explore various 2-D shapes • Identify, name, describe and classify 2-D shapes (rectangle, square, circle and triangle) • Draw and manipulate basic 2-D shapes using ICT
Measure	***Length (Arbitrary units)*** ***Mass (Arbitrary units)*** ***Capacity (Arbitrary units)*** ***Money (Notes and coins up to Rs 100)*** ***Time (To the hour; Days, months)***	• Compare, estimate and measure length of objects using arbitrary units • Recognise and name coins and notes up to Rs 100 • Decompose coins and notes up to Rs 100 • Use coins and notes up to Rs 100 in shopping situations • Tell time to the hour from a clock face • Name days of the week and months of the year
Graphs	***Pictograms***	• Use, read and interpret pictograms • Use ICT to make pictograms representing real-life situations

CONTENT AREAS AND SPECIFIC LEARNING OUTCOMES OF MATHEMATICS FOR GRADE 4

Content Areas	Topics/Sub-topics	By the end of Grade 4, learners should be able to:
Numbers	**Whole numbers** **Numerals and number names** ***(0-10000)*** **Place value** ***(Units, tens, hundreds, thousands, ten thousand)*** **Addition** ***(up to 4-digits with and without carrying)*** **Subtraction** ***(up to 4-digits with and without borrowing)*** **Multiplication** ***(Repeated addition and arrays)*** **Division** ***(sharing and grouping)*** **Number patterns** **Types of numbers** ***(Odd, even)*** **Fractions** ***(Up to tenths; Comparison; Addition (with same and different denominators; Subtraction (with same and different denominators)***	• Recognise, read and write the numbers 0 to 10 000 in numerals and in words • Count, compare and order numbers up to 10 000 • Illustrate the meaning of place value up to 10 000 • Add numbers (up to 4 digits) with and without carrying • Subtract numbers (up to 4 digits) with and without borrowing • Use repeated addition and arrays as multiplication • Use share and repeated subtraction (grouping) as division • Solve two-step word problems involving the 4 operations • Perform mental calculation involving the 4 arithmetic operations • Recognise patterns in number sequences up to 10 000 • Identify odd and even numbers • Write numbers from 1 to 10 000 and perform the 4 arithmetic operations using ICT • Read, write and represent fractions (up to tenths) • Compare and order fractions • Add and subtract fractions with same and different denominators
Geometry	**2-D shapes** **3-D shapes** **Lines** ***(Parallel, Horizontal, Vertical, Inclined)*** **Angles** ***(Compare)***	• Identify, name and describe 2-D shapes • Identify and name 3-D shapes (cube, cuboid, cone, cylinder and sphere) • Recognise parallel, horizontal, vertical and inclined lines • Recognise, name and compare angles • Draw 3-D shapes using ICT
Measure	**Time** ***(Minute-hand and hour-hand)*** **Length** ***(Conversion m/cm)*** **Mass** ***(Conversion kg/g)*** **Capacity** ***(Conversion L/cL)***	• Tell time and draw minute-hand and hour-hand on the clock face to show time • Convert units : length (m to cm), mass (kg to g) and capacity (L to cL) and vice versa • Solve simple word problems involving length, mass, capacity and money
Graphs	**Pictograms** **Bar Chart**	• Collect, organise, represent and interpret data (pictogram and bar chart)

CONTENT AREAS AND SPECIFIC LEARNING OUTCOMES OF MATHEMATICS FOR GRADE 5

Content Areas	Topics/Sub-topics	By the end of Grade 5, learners should be able to:
Numbers	**Whole Numbers**	
	Numerals and number names ***(0-100000)***	• Recognise, read and write numbers up to 100 000 in numerals and in words.
		• Count, compare and order numbers up to 100 000.
	Place value ***(Units, tens, hundreds, thousands, ten thousands, one hundred thousand)***	• Illustrate the meaning of place value of a digit in a given number
		• Use expanded notation for numbers up to 5 digits
	Addition ***(up to 5-digits with and without carrying)*** **Subtraction** ***(up to 5-digits with and without borrowing)***	• Perform addition and subtraction involving up to 5-digit numbers
	Multiplication ***(4-digit by 1-digit numbers; 3-digit by 2-digit numbers; a number by a 2-digit number)***	• Multiply up to 4-digit numbers by 1-digit number.
		• Multiply up to 3-digit numbers by multiples of 10, 100 and 1000, 10000 (product not exceeding 100000)
		• Multiply up to 3-digit numbers by 2-digit numbers
	Division ***(numbers by 10, 100, 1000; number by a 2-digit number)***	• Divide numbers by 10, 100 and 1 000
		• Divide a number by a 2-digit number
		• Solve word problems involving the four arithmetic operations
		• Write numbers from 0 to 100 000 and perform the 4 arithmetic operations using ICT.
	Number patterns	• Recognise and complete patterns in number sequences up to 100 000
	Types of numbers ***(square)***	• Identify square numbers
	Fractions ***(Equivalent; addition/ subtraction (mixed/proper); Multiplication/Division of fractions by whole numbers; convert fractions into decimals)***	• Recognise and use equivalent fractions • Convert mixed numbers to improper fractions and vice versa • Add and subtract fractions (proper and mixed numbers) • Multiply and divide fractions by whole numbers
	Decimals ***(Addition; subtraction; Convert decimals into fractions)***	• Convert fractions into decimals and vice versa • Add and subtract decimals • Solve word problems involving fractions
	Number patterns and sequence ***(Whole numbers, fractions)***	• Recognise and complete patterns in number sequences (up to 100000 and including simple fractions)
	Powers ***(powers of two and three; addition, subtraction, multiplication and division)***	• Recognise number in power of two and three • Convert a number to power form (power 2 and 3 only) • Expand and simplify numbers expressed in power form • Add, subtract, multiply and divide using expansion of powers

	Ratio and proportion ***(Simplest ratio, Ratio as fraction and vice versa; direct proportion***	• Compare two quantities and write down their ratios • Express a ratio in its simplest form • Express a ratio as a fraction and vice versa • Solve simple word problems on ratios • Appreciate the concept of direct proportion • Solve simple word problems on direct proportion
	Average	• Find the average of numbers • Find average from a given sum • Find the sum from a given average • Solve word problems involving averages
Geometry	**Symmetry** ***(one line of symmetry)***	• Identify symmetrical figures • Recognise and draw line(s) of symmetry of figures • Complete symmetrical figures
	Angles **(Right angle, comparison; clockwise/anticlockwise directions)**	• Recognise a right angle • Compare angles • Distinguish between clockwise and anti-clockwise directions
	Shapes ***(Triangles, Quadrilaterals, Pentagon)***	• Recognise and name shapes (up to pentagon) • Identify and name isosceles and equilateral triangles • Identify and name right-angled triangle • Recognise quadrilaterals (square, rectangle, parallelogram, kite) • Name quadrilaterals (square, rectangle, parallelogram, kite) • Draw quadrilaterals (square, rectangle, parallelogram, kite)
Measure	**Length and perimeter** **(*m/cm in practical situations; perimeter of triangles, squares, rectangles)***	• Use the units: metre and centimetre in practical situations • Convert m to cm and vice versa • Perform the four fundamental operations on lengths • Find perimeters of triangles, squares and rectangles • Solve word problems involving lengths and perimeters
	Capacity ***(L, cL, mL)***	• Use the units: millilitre, centilitre, and litre in practical situations • Convert units of volume/capacity from one to another • Perform the four fundamental operations on capacities/volumes • Solve word problems involving capacities

	Mass ***(use of kg/g in practical situations)***	• Use the units: gram and kilogram in practical situations • Convert g to kg and vice versa • Perform the four fundamental operations on masses • Solve word problems involving masses
	Money ***(Profit and Loss)***	• Use the units of currency: Rupees (Rs) and cents (c) in practical situations • Work out shopping problems • Solve word problems involving sharing, profit and loss
	Time ***(Hour, minute, second)***	• Use the units of time: hour, minute and second in practical situations • Convert units of time from one to another • Perform the four fundamental operations on time • Solve simple word problems on time
	Area ***(Triangle, rectangle, square)***	• Find areas of triangles, squares and rectangles • Find width/length of a rectangle given its area • Find the length of one side of square given its area • Perform the four fundamental operations on problems involving areas • Solve word problems involving area
Graphs	**Pictograms**	• Use and interpret pictograms • Draw pictograms
	Bar chart	• Use and interpret bar charts • Draw bar charts using ICT

CONTENT AREAS AND SPECIFIC LEARNING OUTCOMES OF MATHEMATICS FOR GRADE 6

Content Areas	Topics/Sub-topics	By the end of Grade 6, learners should be able to:
Numbers	**Whole numbers**	
	Numerals and number names (0-1000000)	• Recognise, read and write numbers up to 1 000 000 in numerals and in words
	Place value ***(Units, tens, hundreds, thousands, ten thousands, hundred thousands, one million)***	• Illustrate the meaning of place value of a digit in a given number • Use expanded notation for numbers up to 6 digits
	Addition (***up to 6-digits with and without carrying)*** **Subtraction** ***(up to 6-digits with and without borrowing)***	• Perform addition and subtraction involving up to 6-digit numbers
	Multiplication ***(numbers by a 2-gigit number, product not exceeding 100000)***	• Multiply up to 5-digit numbers by 1-digit number • Multiply up to 4-digit numbers by 2-digit numbers. (Product should not exceed 1000000)
	Division ***(up to 6-digit numbers by 1 and 2-digit numbers)***	• Divide numbers by 10, 100, 1 000 and 10000 • Divide 6-digit numbers by 1 and 2 digit numbers • Solve word problems involving the four arithmetic operations • Write numbers from 0 to 100 000 and perform the 4 arithmetic operations using ICT
	Types of numbers ***(prime, composite)***	• Identify prime and composite numbers • Add and subtract simple fractions and mixed numbers
	Fractions ***(Multiply and divide a fraction by another fraction)***	• Multiply and divide a fraction by a fraction
	Decimals ***(Addition, Subtraction, Multiplication; conversion of decimals into fractions and vice versa)***	• Add and subtract decimals in context • Multiply decimals by a 1-digit and 2-digit number to 3 decimal places • Use fractions and decimals in context • Solve word problems involving decimals
	Number patterns ***(whole numbers, fractions, decimals)***	• Identify and use patterns involving whole numbers, fractions, mixed numbers and decimals
	Factors and Multiples	• Find factors and multiples of numbers
	H.C.F. and L.C.M. ***(2 numbers)***	• Find the HCF and LCM of two numbers
	Ratio, Average and Percentage	• Use ratio, average and percentage in context • Solve simple routine and non-routine word problems involving numbers

Geometry	**2-D Shapes** **3-D Shapes** **Symmetry** ***(1-2 lines of symmetry)*** **Angles**	• Identify, name 2-D shapes (up to hexagon) and use them in finding perimeter and area • Identify and name 3-D shapes (Cube, cuboid, cylinder, prism, pyramid) and use Cube/cuboid in finding total surface area and volume • Calculate the number of faces, vertices and edges in each of the 3-D shapes • Draw line(s) of symmetry of a given shape • Complete a figure given one/two lines of symmetry • Find unknown angles • Solve simple routine and non-routine word problems involving geometry
Measure	**Length** ***(km, m, cm, mm)*** **Mass *(kg, g, tonnes)*** **Capacity *(L, cL, mL)*** **Money** ***(Foreign currency; Addition, Subtraction; Percentage Profit, Percentage loss)*** **Area, Total surface area and Volume** **Time** ***(12-hour; 24-hour; GMT)*** **Speed**	• Measure and convert length using km, m, cm and mm • Measure and convert mass using kg, g, t (tonnes to kg and vice versa) and capacity using L, cL, mL • Convert Mauritian currency into foreign currencies (euro, dollar, pound sterling) and vice versa in context Solve problems involving (percentage) profit and (percentage) loss in context • Find areas and perimeters of plane geometric shapes • Find total surface area and volume of cube and cuboid • Use time notation (a.m. and p.m.) and perform addition and subtraction of time (second, minutes, hours, days, years) • Read and write time using 12-hour and 24-hour clock • Relate international time to G.M.T. • Use basic notion of speed in context • Solve simple routine and non-routine word problems measure (using ICT)
Graphs	**Pictogram** **Bar chart** **Pie chart** **Line graph** **Coordinates**	• Read and interpret data represented in charts (Pictogram, Bar Chart, pie chart, line graph) • Find and use coordinates in the x-y plane • Solve simple routine and non-routine word problems involving graphs

ACHIEVEMENT CRITERIA AND LEVELS (GRADE 1)

Achievement criteria are assessment objectives that are scaffolded and represent clear measurable outcomes. The levels reflect the competencies in terms of knowledge, skills, values and attitudes expected of pupils. The criteria move progressively from Level 1 (basic competency) to Level 3.

CONTENT AREA, SPECIFIC LEARNING OUTCOMES, ACHIEVEMENT CRITERIA AND LEVELS FOR GRADE 1

	Level 1 **Knowledge and Comprehension**	**Level 2** **Application**	**Level 3** **Analysis**
Numbers	• Recognise numbers up to 10 in numerals • Count numbers up to 10 • Recognise patterns in number sequences	• Read and write numbers up to 10 in numerals and words • Compare and order numbers up to 10 • Perform addition of two numbers (sum not exceeding 10) • Complete patterns in number sequences • Perform and explain mental calculations involving addition • Write numbers 0-10 and perform addition (sum not exceeding 10) using ICT	
Geometry	• Recognise and name colours and shapes • Identify patterns involving 2-D shapes	• Sort out objects according to one attribute and state the sorting rule • Compare shapes in the environment • Complete patterns (2-D shapes) • Draw and manipulate basic 2-D shapes using ICT	
Measure	• Recognise and name coins up to Rs 10	• Decompose coins up to Rs 10 • Sort out and compare lengths of objects • Sort out and compare masses of objects • Distinguish between night and day	• Use coins up to Rs 10 in shopping situations

CONTENT AREA, SPECIFIC LEARNING OUTCOMES, ACHIEVEMENT CRITERIA AND LEVELS FOR GRADE 2

	Level 1 **Knowledge and Comprehension**	**Level 2** **Application**	**Level 3** **Analysis**
Numbers	• Recognise numbers up to 100 in numerals • Count numbers up to 100 • Recognise place value for numerals to 10 • Recognise patterns in number sequences	• Read and write numbers up to 100 in numerals and words • Compare and order numbers up to 100 • Perform addition of two numbers (not involving carrying and sum not exceeding 100) • Perform subtraction of two numbers (not involving borrowing) • Multiply a number (from 0 to 10) by 2 (perform multiplication as a repeated addition, with and without manipulatives) • Divide numbers up to 20 by 2 (sharing equally) • Complete patterns in number sequences • Perform mental and explain calculations involving addition and subtraction	• Solve missing value problems • Solve simple word problems involving the 4 arithmetic operations

Geometry	• Identify and complete patterns using 2-D shapes	• Draw and manipulate basic 2-D shapes using ICT • Sort out objects according to two attributes	• Explain the sorting rule (up to two attributes)
Measure	• Measure length of objects using arbitrary units • Measure mass of objects using arbitrary units • Measure volume of objects using arbitrary units • Recognise and name coins and notes up to Rs 100 • Tell time to the hour from a clock face • Name days of the week and months of the year.	• Decompose an amount into coins and notes (up to Rs 100)	• Use coins and notes up to Rs 100 in shopping situations

CONTENT AREA, SPECIFIC LEARNING OUTCOMES, ACHIEVEMENT CRITERIA AND LEVELS FOR GRADE 3

	Level 1 **Knowledge and Comprehension**	**Level 2** **Application**	**Level 3** **Analysis**
Numbers	• Recognise, read and write numbers up to 1 000 in numerals and in words • Recognise and use ordinal numbers (1st to 20th) • Use unit fractions as part of a whole • Read, write and represent simple fractions (e.g. 1/2, 1/3, 1/4) • Recognise the roman numerals up to 20	• Place value • Count, compare and order numbers up to 1000 • Add 2 numbers (up to 3 digits) without/with carrying • Subtract 2 numbers (up to 3 digits) without/with borrowing • Multiply a number (from 0 to 10) by 2, 3, 4 and 5 • Divide a number (multiple of 2, 3, 5 and 5) by 2, 3, 4 and 5 respectively • Solve one-step simple/routine word problem involving the 4 operations • Perform mental calculations involving the 4 operations • Read and write roman numerals up to 20 • Use fractions in real-life situations • Use of ICT for representation and computation of numbers	• Identify patterns in number sequences up to 1000 and explain the rule • Explain mental calculations involving the 4 operations • Solve word problem involving the 4 operations

Geometry	• Manipulate, sort, and represent various 2-D shapes • Identify and name 2-D shapes (rectangle, square, circle and triangle)	• Explore, describe and classify 2-D shapes (rectangle, square, triangle and circle) • Draw and manipulate basic 2-D shapes using ICT	• Construct and extend shape patterns
Measure	• Measure length (m/cm), mass (kg/g) and capacity (L/cL) • Tell time to the hour • Count amount of money in rupees and cents up to Rs 1000	• Perform arithmetic operations involving length, mass, capacity and money • Tell time to the half hour/ quarter hour • Use of 'a.m.' and 'p.m.' in real-life situations	• Solve word problems involving length, mass, capacity and money
Graphs	• Recognise pictograms	• Draw pictograms	• Interpret pictograms • Use ICT to draw and interpret pictograms in real life situations

CONTENT AREA, SPECIFIC LEARNING OUTCOMES, ACHIEVEMENT CRITERIA AND LEVELS FOR GRADE 4

	Level 1 **Knowledge and Comprehension**	**Level 2** **Application**	**Level 3** **Analysis**
Numbers	• Recognise numbers up to 10000 in numerals • Read numbers up to 10000 in numerals • Write the numbers up to 10000 in numerals • Count numbers up to 10000 • Recognise and read numbers up to 10000 in words • Identify even and odd numbers • Identify fractions from pictorial representations	• Write the numbers up to 10000 in words • Identify and complete patterns in number sequences up to 10000 • Compare numbers up to 10000 • Order numbers up to 10000 • Add numbers (up to 4 digits) without/with carrying • Subtract numbers (up to 4 digits) without/with borrowing • Interpret multiplication as repeated addition • Interpret multiplication in terms of arrays • Division (sharing and grouping) • Perform and explain mental calculation involving the 4 arithmetic operations • Interpret fractions in symbolic forms • Add and subtract fractions with same denominators • Use of ICT for representation in computation of numbers, including fractions	• Solve two-step word problems involving the 4 operations • Order and compare fractions • Add and subtract fractions with different denominators

Geometry	• Identify and name 2-D shapes • Identify and name 3-D solids (cube, cuboid, cone, cylinder and sphere) • Recognise horizontal, vertical, inclined and perpendicular lines	• Describe 2-D shapes • Describe 3-D shapes (cube, cuboid, cone, cylinder and sphere) • Name and compare angles • Use of arbitary unit to compare angles • Draw 3-D shapes using ICT	• Analyse relevant geometric properties in relatively complex figures
Measure	• Read and write units of measure (time, length, mass, capacity and money)	• Draw minute-hand and hour-hand on the clock face to show time; interpret a calendar • Convert units : length (m to cm) and vice versa • Convert units : mass (kg to g) and vice versa • Convert units : capacity (L to cL) and vice versa • Convert units : time (h to min) and vice versa	• Solve word problems involving length • Solve word problems involving mass • Solve word problems involving capacity • Solve word problems involving money • Solve word problems involving time
Graphs	Collect data	• Organise data	• Represent data (pictogram and bar chart) • Interpret data (including pictogram and bar chart)

CONTENT AREA, SPECIFIC LEARNING OUTCOMES, ACHIEVEMENT CRITERIA AND LEVELS FOR GRADE 5

	Level 1 **Knowledge and Comprehension**	**Level 2** **Application**	**Level 3** **Analysis**
Numbers	• Recognise numbers up to 100000 in numerals and in words • State the place value of a digit in a given numbers. • Count up to 100000 • Recognise patterns in number sequences (up to 100000 and including simple fractions) • Recognise and use equivalent fractions • Identify square numbers	• Read and write numbers up to 100000 in numerals and in words • Use expanded notation for numbers up to 5-digit • Order and compare numbers up to 100000 • Perform addition and subtraction involving up to 5-digit numbers • Multiply up to 3-digit numbers by 2-digit numbers • Multiply up to 4-digit numbers by 1-digit numbers. • Multiply up to 3-digit numbers by multiples of 10, 100 , 1 000 and 10000 (product not exceeding 100000) • Divide a number by a 2-digit number • Divide numbers by 10, 100 and 1 000 • Identify and complete a number pattern • Complete a sequence by adding/subtracting • Complete a sequence using the same multiplier/divisor • Compare two quantities and write down their ratios • Express a ratio as a fraction and vice versa • Use the concept of direct proportion • Recognise number in powers of two and three • Express a number in power form (powers 2 and 3 only). • Convert mixed numbers to improper fractions and vice versa • Add and subtract fractions (proper and mixed numbers) • Multiply and divide fractions by whole numbers • Add decimals (up to 2 decimals places) • Subtract decimals (up to 2 decimals places) • Convert fractions into decimals and vice versa • Express a ratio in its simplest form • Compute averages	• Solve word problems on fractions • Solve simple word problems on ratios • Solve simple word problems on direct proportion • Solve word problems involving averages

Geometry	• Identify symmetrical figures. • Recognise line(s) of symmetry of figures • Recognise a right angle • Distinguish between clockwise and anti-clockwise directions • Recognise and name shapes (up to pentagon) • Identify isosceles and equilateral triangles • Identify right-angled triangle. • Recognise quadrilaterals (square, rectangle, parallelogram, kite)	• Draw line(s) of symmetry of figures • Compare angles	• Complete symmetrical figures given one line of symmetry • Complete figures to triangles and quadrilaterals
Measure	• Recognise basic units of measurement (m/cm) , (mL/cL), (g/kg) and (Rs/c) and their uses in appropriate situations • Use the units: gram and kilogram in practical situations. • Recognise the use of the units of time: hour, minute and second in practical situations	• Convert m to cm and vice versa. • Perform the four fundamental operations on lengths • Convert units of volume/capacity from one to another. • Perform the four fundamental operations on capacities/volumes • Convert g to kg and vice versa. • Perform the four fundamental operations on masses • Use the units of currency: Rupees (Rs) and cents (c) in practical situations • Work out shopping problems • Convert units of time from one to another • Perform the four fundamental operations on time • Find perimeter of a given shape • Find areas of triangles and quadrilaterals	• Solve word problems involving lengths • Solve word problems involving capacities • Solve word problems involving masses • Solve word problems involving money, including profit and loss. • Solve simple word problems on time • Solve word problems involving perimeter and area
Graphs	• Recognise pictograms • Recognise bar charts	• Draw pictograms • Draw bar charts using ICT	• Interpret pictograms • Interpret bar charts

CONTENT AREA, SPECIFIC LEARNING OUTCOMES, ACHIEVEMENT CRITERIA AND LEVELS FOR GRADE 6

	Level 1 Knowledge and Comprehension	**Level 2 Application**	**Level 3 Analysis**
Numbers	• Recognise, read and write numbers up to 1 000000	• Interpret place value of numbers (including decimals) • Express numbers in the expanded form • Perform the four mathematical operations involving whole numbers, fractions, decimals, ratio and proportion • Identify prime and composite numbers • Find HCF and LCM of two numbers • Use percentages • Identify and use patterns involving fractions, mixed numbers and decimal fractions • Perform and describe simple mental computation	• Use fractions and decimals in context • Use ratio, average and percentage in context • Solve word problems involving the four operations
Geometry	• Identify faces, edges and vertices of 3-D shapes • Identify and name 2-D shapes (up to hexagon) • Identify and name 3-D shapes (cube, cuboid, cylinder, prism, pyramid) • Differentiate among different types of angles (acute, obtuse and reflex) • Recognise scalene triangles	• Work with angles in degrees • Apply geometric properties to solve simple problems	• Complete figures given two lines of symmetry • Apply geometric properties to solve harder problems
Measure	• Recognise basic notion of speed	• Use time notation (a.m. and p.m.) • Relate international time to G.M.T. • Find areas and perimeter of plane geometric figures • Find total surface area and volume of cube and cuboid • Measure mass and capacity in standard units (kg, g, L, cL, mL) • Convert Mauritian currency into foreign currencies (euro, dollar, pound, sterling) vice versa in context • Perform addition and subtraction involving time (seconds, minutes, hours, days, years) • Perform calculations involving speed	• Estimate and compare measures of distance and time • Solve word problems involving different types of measures (mass, capacity, volume, time) • Solve simple word problems involving speed
Graphs		• Read data represented in charts (Pie Chart)	• Interpret data represented in charts (Pie Chart) • Solve simple routine and non-routine word problems involving graphs (Pictogram, Bar Chart, Pie Chart)

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